WORKED SOLUTION

STEP III 1990, Question 16

Probability and Statistics • Geometric Probability

STEP III 1990, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q16

This question has a full guided treatment in *Beyond STEP*, Section 12.21, *Breaking a Stick*: https://beyondstep.keithmcnulty.org/ch12/broken-stick.html

The key idea

Three lengths with a fixed total make a triangle exactly when each one is less than half the total. Once a triangle exists, it's obtuse exactly when the square of the longest side exceeds the sum of the squares of the other two (the cosine rule). Both parts turn into conditions on the random variables, and the probabilities become lengths of intervals in (i) and areas of regions in (ii).

Solution

Part (i)

The three pieces have lengths X, $\frac{1}{2}(1-X)$ and $\frac{1}{2}(1-X)$, with total 1. Three positive lengths form a triangle if and only if each is less than the sum of the other two, that is, less than half the perimeter, here $\frac{1}{2}$. The two equal pieces satisfy $\frac{1}{2}(1-X) < \frac{1}{2}$ whenever $X > 0$, so the only condition is $X < \frac{1}{2}$. Let T be the event that a triangle can be formed. Then

$$\mathrm{P}(T) = \mathrm{P}\left(0 < X < \tfrac{1}{2}\right) = \tfrac{1}{2}.$$

A triangle is obtuse if and only if the square of its longest side is greater than the sum of the squares of the other two. Which side is longest depends on X.

- If $X \leq \frac{1}{3}$, then $X \leq \frac{1}{2}(1-X)$, so one of the equal pieces is a longest side. The test would be $\frac{1}{4}(1-X)^2 > \frac{1}{4}(1-X)^2 + X^2$, i.e. $X^2 < 0$, which never happens. The triangle is isosceles with its base no longer than the equal sides, so it's never obtuse.

- If $\frac{1}{3} < X < \frac{1}{2}$, the longest side is X, and the triangle is obtuse if and only if

$$X^2 > 2 \cdot \tfrac{1}{4}(1-X)^2 \iff X > \frac{1-X}{\sqrt{2}},$$

taking positive square roots, since both sides are positive. This rearranges to $X(\sqrt{2}+1) > 1$, i.e. $X > \dfrac{1}{\sqrt{2}+1} = \sqrt{2}-1 \approx 0.414$, which does lie in $\left(\frac{1}{3}, \frac{1}{2}\right)$.

Let O be the event that the triangle is obtuse. Then $O \cap T$ is the event $\sqrt{2}-1 < X < \frac{1}{2}$, and since X is uniform,

$$\mathrm{P}(O \mid T) = \frac{\mathrm{P}(\sqrt{2}-1 < X < \frac{1}{2})}{\mathrm{P}(T)} = \frac{\frac{1}{2} - (\sqrt{2}-1)}{\frac{1}{2}} = 1 - 2\sqrt{2} + 2 = 3 - 2\sqrt{2},$$

as required.

Part (ii)

The construction is always possible. The triangle R with vertices $(1,0)$, $(1,1)$, $(0,1)$ is the set of points with $x \le 1$, $y \le 1$ and $x + y \ge 1$. The proposed sides x, y, $2-x-y$ have total 2, and they form a (possibly degenerate) triangle if and only if all three are non-negative and none exceeds half the perimeter, 1:

- $0 \le x \le 1$ and $0 \le y \le 1$ hold on R;
- $2 - x - y \le 1$ is just $x + y \ge 1$, and $2 - x - y \ge 0$ follows from $x \le 1$, $y \le 1$.

So every point of R gives a triangle. It is degenerate only on the edges of R, which have probability zero.

When is $\angle ABC$ obtuse? The angle at B lies between BA and BC, so the side opposite it is $CA = y$. By the cosine rule, $\angle ABC$ is obtuse if and only if $CA^2 > AB^2 + BC^2$:

$$y^2 > x^2 + (2-x-y)^2.$$

Expanding $(2-x-y)^2 = (2-x)^2 - 2(2-x)y + y^2$, the y^2 terms cancel, and the condition becomes

$$2(2-x)y > x^2 + (2-x)^2 = 2x^2 - 4x + 4.$$

Since $x \le 1$, we have $2 - x > 0$, so we can divide by $2(2-x)$ without reversing the inequality:

$$y > \frac{x^2 - 2x + 2}{2 - x},$$

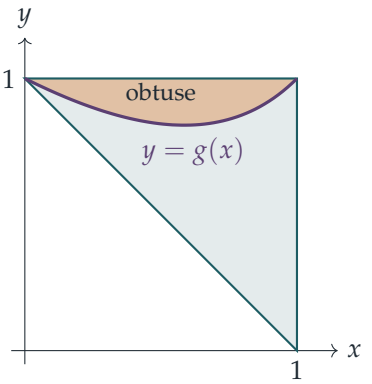
as required. Call the right-hand side $g(x)$.

Ask yourself Does the curve $y = g(x)$ actually stay inside the triangle R?

We need it to lie between the lines $y = 1 - x$ and $y = 1$ for $0 \le x \le 1$, so that the obtuse region is simply the part of R above the curve. Indeed,

$$1 - g(x) = \frac{(2-x) - (x^2 - 2x + 2)}{2 - x} = \frac{x(1-x)}{2-x} \ge 0, \qquad g(x) - (1-x) = \frac{x}{2-x} \ge 0,$$

with $g(0) = g(1) = 1$. So the curve runs from $(0,1)$ to $(1,1)$, dipping below $y = 1$ but never below the hypotenuse $x + y = 1$.



The probability. The triangle R has area $\frac{1}{2}$, so the joint density is 2 on R, and

$$\mathrm{P}(\angle ABC \text{ obtuse}) = 2 \times \text{area}\{(x,y) : g(x) < y \le 1,\ 0 \le x \le 1\} = 2\int_0^1 (1 - g(x))\,\mathrm{d}x.$$

To integrate g, substitute $u = 2 - x$, so $x = 2 - u$ and

$$x^2 - 2x + 2 = (2-u)^2 - 2(2-u) + 2 = u^2 - 2u + 2, \qquad g = \frac{u^2 - 2u + 2}{u} = u - 2 + \frac{2}{u}.$$

As x runs from 0 to 1, u runs from 2 to 1, and $\mathrm{d}x = -\mathrm{d}u$, so

$$\begin{aligned}\int_0^1 g(x)\,\mathrm{d}x &= \int_1^2 \left(u - 2 + \frac{2}{u}\right)\mathrm{d}u = \left[\tfrac{1}{2}u^2 - 2u + 2\ln u\right]_1^2 \\ &= (2 - 4 + 2\ln 2) - \left(\tfrac{1}{2} - 2\right) = 2\ln 2 - \tfrac{1}{2}.\end{aligned}$$

Hence

$$\mathrm{P}(\angle ABC \text{ obtuse}) = 2\left(1 - 2\ln 2 + \tfrac{1}{2}\right) = 3 - 4\ln 2,$$

as required.

Checks

- Numerically, $3 - 2\sqrt{2} \approx 0.1716$ and $3 - 4\ln 2 \approx 0.2274$. A Monte Carlo simulation with 10^6 trials gave about 0.1719 and 0.2278.

- Both answers lie between 0 and 1. In (ii), the curve touches $y = 1$ at both ends: at $x = 0$ the side BC vanishes, and at $x = 1$ the side AB vanishes, so both are degenerate cases, as you'd expect.
- A Simpson's-rule evaluation of $\int_0^1 g(x)\,\mathrm{d}x$ gives 0.886294, matching $2\ln 2 - \frac{1}{2}$.

Common slips

- In (i), forgetting to condition: the unconditional probability of an obtuse triangle is $\frac{3}{2} - \sqrt{2}$, half the required answer.
- In (ii), using the wrong side in the cosine rule. The angle at B is opposite $CA = y$, not AB.
- Forgetting that the density on R is 2, not 1, which gives an answer half the size.

Beyond STEP

Turning a probability into a length or an area is *geometric probability*, and breaking a stick at random is one of its classic problems: break a stick at two independent uniform points and the chance of a triangle is $\frac{1}{4}$. Part (ii) is a first look at continuous joint distributions, where probabilities are double integrals of a joint density over a region. Grimmett and Welsh, *Probability: An Introduction*, treats joint densities and has several problems of this kind.

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